



Port performance indicators construction based on the AIS-generated trajectory segmentation and classification

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Abstract

With coverage of worldwide shipping routes and straightforward accessibility, vessels' navigational data from AIS have emerged as a potential leading source of knowledge for maritime stakeholders. However, transforming the raw AIS messages into meaningful indicators requires comprehensive work that involves enrichment with multiple relevant data sources. In this study, a fine-grained technique to process AIS tracks for the purpose of constructing port performance indicators is proposed. The technique involves trajectory segmentation to separate the stopping and underway segments of AIS-generated trajectories and trajectory classification to identify moored and at-anchor vessels from the stopping segments. With the fusion of relevant information, the method estimates the port boundary and identifies the associated berthing and anchorage locations to estimate turnaround time and construct port connectivity indicators. AIS datasets recorded over eight months covering the waters of Indonesia, Malaysia, and Singapore, known as one of the world's busiest shipping routes, are selected to evaluate the proposed method. The trajectory segmentation demonstrated 98–99% accuracy, while the classification achieved 95–97%. The estimation of the vessel turnaround time closely matched the UNCTAD data with an error rate of 2.9%. These results prove the proposed approach's practicality in contributing to the real-world scenario.

Keywords Automatic identification system · Port performance indicators · Vessel turnaround time · Stopping vessel detection · Vessel trajectory classification · Port boundary estimation

1 Introduction

Delivering goods on seagoing vessels is a highly economical, energy-efficient, and reliable long-distance transportation method, accounting for more than 80 percent of the worldwide merchandise trade volume. A notable and increasing share of this quantity, comprising approximately 35 percent of the overall volume and exceeding 60 percent of the total commercial worth, is transported within containers [1]. Indeed, container shipping is widely recognized as a global industry and a catalyst for globalization [2].

To ensure efficiency in maritime transport, consistent and reliable port performance has been indispensable. The adverse impacts of inadequate performance at a port can extend beyond its immediate surroundings to affect other ports. Poor performance at a single port has the potential to disrupt the entire itinerary of the shipping route [1]. However, constructing indicators to assess port performance has proven to be complicated. The challenge stems from various factors, such as the quality, consistency, and accessibility of data collected by ports, differing definitions used, and organizations' willingness to share data. These obstacles have impeded the development of consistent indicators for comparing port performance across different ports and time [1]. The ability to quantitatively measure and compare port performance is crucial in designing and optimizing port operations and container logistics [3].

One of the most important and commonly accepted indicators of port performance is vessel turnaround time, the time vessels spend in port [2, 4–6]. The reduction in the vessel turnaround time is a crucial factor contributing to the success of containerization. From the perspective of

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supply chain managers, port authorities, policymakers, and other stakeholders, information on turnaround time is useful for planning infrastructure and improving port operations. Meanwhile, from the perspective of shipping carriers, ship time in port is a cost factor, and minimizing terminal costs is an objective of service design [2].

Despite its significance, measuring the duration ships spend in ports has been challenging [2, 6]. It is difficult to acquire the actual time metrics directly from individual ports due to specific reasons, such as confidentiality concerns [2]. In addition, works that predominantly rely on statistical data provided by port authorities and shipping companies to assess port performance typically involve huge human effort and are time-consuming. Due to the delay and absence of relevant information, these works sometimes yield sub-optimal and incomplete outcomes [3, 5]. Thus, the Automatic Identification System (AIS) has been utilized as an alternative data source.

The development of AIS was initiated in the 1990s for the primary purposes of preventing collisions and improving navigational safety [7, 8]. It was implemented in the 2000s when the International Maritime Organization (IMO) adopted a new requirement for ships to carry AIS [8–10]. Ships involved in international voyages with a gross tonnage (GT) of 300 or more, cargo ships not engaged in international voyages of 500 GT or more, and all passenger ships, regardless of size, are required to be fitted with AIS device [11–13].

AIS enables the sharing of navigational data among vessels, terrestrial base stations, and/or satellites. It transceives static, dynamic, and voyage-related information, such as ship name, type, MMSI, position, position timestamp, speed over ground (SOG), course over ground (COG), heading, navigational status, destination, estimated time of arrival, and draught. MMSI stands for Maritime Mobile Service Identity. It is a unique nine-digit number used by maritime digital selective calling (DSC), AIS, and certain other equipment to uniquely identify a ship or a coast radio station [14]. The navigational status indicates the movement state of the vessel, such as moored, at anchor, underway, aground, restricted maneuverability, constrained by draught, not under command, and engaged in fishing. Since it is manually set by the ship's crew [13], the reported navigational status may differ from the actual status due to human error [15]. AIS device transmits messages containing the information every 2 to 10 s for ships moving faster than 3 knots and every 3 min when they are at anchor or moored and not moving faster than 3 knots [13].

With coverage of worldwide shipping routes and straightforward accessibility, vessels' navigational data from AIS have emerged as the potential leading source of information on maritime transport. The World Bank, the UNCTAD (United Nations Conference on Trade and Development),

the IMF (International Monetary Fund), the ADB (Asian Development Bank), and experts of various organizations have utilized AIS as a maritime data source to derive knowledge from the traffics of seagoing vessels. A comprehensive review regarding the applications of AIS data for solving a broad range of maritime problems was conducted by Yang et al. [7]. The number of studies on AIS data is generally more than doubled every three years. However, amongst 171 literature, only one study proposed a method to derive port performance indicators.

This paper is motivated by the contradicting reality that despite the importance of the port performance indicators, and the extensive exploration of the promising AIS based maritime data, study on exploiting the AIS data to construct port performance indicators is scarce. Understanding the significance of these indicators in the maritime industry, we aim to fill this gap in the literature.

The construction of port performance indicators involves techniques for identifying ship-stopping events within the area of the port of interest (POI) [3–5]. The techniques shall need to complete the following tasks: (1) estimate the port area, (2) identify the berthing location and anchorage area of the port, (3) identify ship stopping event, and (4) classify the stopping event into moored and at-anchor stops. However, to the best of our knowledge, no existing study has provided a thorough method to accomplish the four tasks with evaluations by exploiting AIS data to construct port performance indicators.

This study identifies the following issues that remain in the existing works related to the four tasks. First, the study on extracting stopping tracks within port area does not provide a method for identifying port boundaries [16]. Second, the paper that proposed techniques to predict port boundary does not elaborate on its work to identify the port's berthing location and anchorage area [5], which is required in the process of constructing port performance indicators. Third, the research on constructing trade volume indicators proposes a method of identifying port boundaries, including a technique for recognizing vessels stopping tracks. However, neither the stopping duration nor the stop mode is specified [17].

Thus, this paper proposes a comprehensive technique to estimate port boundaries, recognize the berthing location and anchorage area associated with the POI, and construct port performance indicators based on AIS-generated vessel trajectories. The technique answers the question of how to transform raw AIS data into meaningful values for constructing port performance indicators. First, a segmentation procedure is applied to separate the stopping and underway segments of each vessel trajectory. Underway means that a vessel is not at anchor, or made fast to the shore, or aground [18]. Stopping indicates that a vessel is either at anchor, or made fast to the shore, or aground. The stopping segments are utilized to estimate port boundary. Second, the stopping seg-

ments are classified into moored and at-anchor segments. The moored segments represent berthing vessels, while the at-anchor segments correspond to anchoring vessels. The port's berthing and anchorage locations are predicted with moored and at-anchor segments, respectively. The relation between them is examined with the underway segments. Third, the port performance indicators are constructed by integrating the trajectory segmentation and classification results with the vessel database accessible at Balticshipping.com [19].

This study aims to discover the events and areas of interest described in the ADB's work [5] to extract knowledge for constructing port performance indicators by utilizing transparently accessible resources, such as AIS data, geoBoundaries [20], world port index [21], and vessel database [19].

The method and results presented in this paper are expected to contribute in three ways. First, provide an accurate technique to separate stopping and underway segments from a vessel trajectory. The stopping segments can be used to analyze vessel activities within port areas, while the underway segments are used to comprehend vessel movement patterns at sea to identify anomalous behavior. Second, identify the *de facto* boundary and zoning of the POI inferred from the real-world vessel activities within the area. Maritime space management can utilize the estimation to evaluate the compatibility between activities observed and current zoning regulations [22]. Third, contribute to the world's maritime industries by proposing a sensible and transparent approach to constructing port performance indicators. The remainder of this paper is organized as follows. Section 2 reviews the existing works relevant to this study. Section 3 describes the proposed approach. Then, the experiment results and evaluation are discussed in Sect. 4. Finally, Sect. 5 concludes the content of this paper and specifies future works.

2 Related work

The UNCTAD has created a series of key indicators to measure port performance, such as the Number of Port Calls,¹ Turnaround Time, Port Waiting Time, Cargo Handling, and Liner Shipping Connectivity [25, 26]. The Liner Shipping Connectivity is defined based on the number of port calls, their container carrying capacity, the number of services and companies, the size of the largest ship, and the number of other countries connected through direct liner shipping services [27].

The vessel turnaround time, which is the time vessels spend in port [2], is adopted by the World Bank to assess

port performance and published The Container Port Performance Index accordingly [1]. Since the turnaround time is influenced by ship arrival rates, waiting times, and cargo off-take, it can be a good indicator of the overall port efficiency [25]. Slack et al. concluded the importance of the vessel turnaround time in port operations based on their review of relevant maritime literature [2]. The review indicates that port performance can influence the time ships spend in port. More efficient ports are probable to turn ships around faster than others despite high volumes. Karnoiji et al. also imply the significance of vessel turnaround time for measuring port performance in their recent study on identifying factors affecting the time [4]. Similarly, Adeyanju and Ojekunle emphasize the same notion in their book published in 2014 [6].

In a more recent publication, the UNCTAD proposes an overarching series of port performance indicators and groups them into seven categories: (1) governance indicators, (2) human resource indicators, (3) financial indicators, (4) resilience indicators, (5) vessel operation indicators, (6) cargo operation indicators, and (7) environmental sustainability indicators [28]. This new proposal classifies vessel turnaround time as time indicators within the category of vessel operation indicators, along with the indicators of vessel characteristics.

Another study on port performance measurement was conducted by Chen et al., which utilizes a different set of key indicators [3]. They measure and compare port performance with four key indicators: (1) Ship Traffic, (2) Container Throughput, (3) Berth Utilization, and (4) Terminal Productivity.

Ship Traffic is defined as the number of ships arriving at a terminal (or a port) for container handling during a period of time. Container Throughput is the number of containers handled at a terminal (or a port) during a period of time. Berth Utilization is the percentage of time that the berths of a terminal (or a port) are occupied by ships during a period of time. Terminal Productivity is the number of containers handled per hour at a terminal over a period of time.

Their method involves stationary event detection using a δd threshold, defined as the distance between two adjacent points. To retrieve only the stationary event of cargo handling and remove other stationary events such as vessel at anchor, repair at shipyard, and temporary stop, they apply a supervised model by training an SVM (Support Vector Machine) classifier with a manually labeled dataset. The classifier uses three features: *Ship-Terminal Distance*, *Heading Direction Variation*, and *Berthing Duration*. They validated the classification method with a manually labeled dataset of 500 ship traces from the Port of Hong Kong, where 70% are used as the training dataset, and 30% are for validation. The classification achieved 0.906 precision and 0.942 recall. They further evaluated the *Container Handling Event Identification* and the *Container Handling Quantity Estimation* for the port of

¹ A port call is a joint process in which many different parties work together to get vessels into the port, operate them, and get them out of the port, safely and efficiently [23]. Or, simply a visit by a ship to a port [24].

Hong Kong and Singapore with Mean Absolute Percentage Error (MAPE) of 0.4–1.64%.

Although the approach achieved practical accuracies in the defined parameters of constructing the intended key performance indicators, their efficacy cannot be evaluated with the more commonly adopted indicators, such as vessel turnaround time. However, the method has one task in common with other studies, which is the task of identifying stopping events.

To measure the time a vessel spends in a port based on the AIS tracks, the stopping tracks of each vessel within the port boundary during the relevant period of time need to be extracted.

A comprehensive study on extracting ship-stopping information from AIS tracks is proposed by Yan et al. [16]. The study further divides the ship-stopping events into two stopping modes: berth and anchorage stops. The method to extract stopping tracks from AIS data employs the integration of trajectory features (speed, duration, and distance between two adjacent tracks) and geographic semantics (water depth, road, coastline) to target ship stoppings in port areas. The experiment on a dataset of 359,322 AIS records reported by 70 vessels proved its effectiveness with 94% accuracy, outperforming the ship-stopping point identification algorithms of previous studies.

After extracting the stopping tracks, the study applied a random-forest-based ship stopping mode classification to classify whether a stopping segment is a berthing or anchoring stop. The technique utilizes 15 ship-stopping mode features selected from trajectory features and geographic semantics. The study's experiment on 202 stopping segments proved the classification practicality with 93% accuracy. However, even though the study targets ship stopping in port areas, neither port boundary identification nor stopping duration calculation is provided.

In line with the work of Yan et al., the ADB published a report on a methodological framework to unlock maritime insights using AIS data [5]. They recognize the drawbacks of the official statistics that traditionally rely on surveys, censuses, and administrative data. These traditional ways often involve time-consuming data collection, processing, and analysis, causing delays in the delivery of relevant information. The delays limit the opportunity of various stakeholders to make real-time decisions or immediate responses to emerging trends. On the other hand, the increasing availability of AIS data has led to a growing interest among researchers in its potential for generating real-time indicators. Thus, they committed to proposing a framework to derive statistical indicators from AIS data that enable near real-time analysis of maritime activity for ports and passages.

The framework consists of two functions: detecting events of interest (EOIs) and identifying areas of interest (AOIs).

EOIs are specific maritime incidents or activities relevant to a specified indicator. AOIs are the geographic locations where such events occur. Consider an indicator of the volume of traffic in the Port of Singapore. The EOIs would include all activities that vessels do in ports, such as loading cargo, refueling, and anchoring. The AOI would be the boundaries of the port. Ships that match this EOI-AOI description provide the data for constructing the indicator.

Three approaches are presented to define the boundary of the POI: manually drawing the boundary, distance-based, and cluster-based. However, the approaches do not involve identifying the port's berthing location and anchorage area, nor do they demonstrate the connection between them. The connection between the two places can be inferred by whether vessel traffic exists. If there is no traffic, the anchorage area is not connected to the berthing location. In other words, it does not belong to the same port as the port of the berthing location.

Three indicators are used as the measure of port activities: (1) the count of unique vessels in a port, (2) the number of arrivals in a port, and (3) the median time spent by vessels in a port, which is regarded as the most influential factor in port efficiency and performance. For the third indicator, the EOIs are the vessel's entry to, halt in, and exit from the port. Its entry and exit mark the start and end of the total time spent in the port. However, this method will also calculate the time spent by vessels in the anchorage area within the port. While at anchor, vessels may conduct activities unrelated to port performance, such as repairs or waiting for orders. This way, the calculation of the median time spent in port may produce misleading results.

The report's experiment results are validated by comparing them against alternative data sources, either AIS-derived or official statistics from port authorities. It demonstrates 3–4% of absolute percentage errors for the number of ship arrivals within the POI. Based on the validation results, this report emphasizes the practicality of AIS-derived indicators as a real-time alternative to official maritime statistics, providing valuable insights remarkably ahead of their official release.

Another study on empowering AIS data to generate economically meaningful indicators of international trade is proposed by Cerderio et al. [17]. They build real-time indicators of world seaborne trade using raw AIS data to provide a more immediate picture of global trade flows than is currently available. The study was triggered by the motivation that businesses and policymakers need to be timely informed when making investment decisions or crafting policy recommendations. Meanwhile, most trade data are available with a one to three-month lag. Their approach fits with official trade statistics for many countries and the world in aggregate [17].

They consider port boundary as the first building block of their trade-nowcasting indicators. Instead of using domain

knowledge to draw port boundaries, they propose an approach to infer them from the AIS data. The approach employs the DBSCAN (Density-based Spatial Clustering of Application with Noise) algorithm [29] to construct geographical clusters of stopping tracks. Each cluster's convex hull is computed, which results in a polygon used to estimate port area. The stopping vessel tracks are identified based on the navigational status information of the AIS messages and a speed threshold. Tracks with a reported navigational status *moored* or *at anchor* and speed less than 0.5 knots are identified as stopping vessels for loading/offloading work. This is based on the fact that loading and offloading work need vessels to stand still and on the assumption that the ship's crews will correctly update the vessel's navigational status.

Having identified the port polygon, they verify every stopping track, whether transmitted by a true port call or just by a passing ship. The stopping tracks within the port polygon are compared with the official port-call data produced by the U.S. Army Corps of Engineers Navigation Data Center (NDC) to look for matches. The IMO (International Maritime Organization) number is used as the key. As a result, 94.3% of port polygon visits, identified as stopping vessels, match the NDC's port calls. Following this, a random forest classifier is trained with the official NDC's port calls using 23 features. This classifier is then applied to their worldwide AIS-derived stopping tracks and creates an estimated global port call database.

The implementation of DBSCAN algorithm to identify stationary points within the AIS messages is also proposed by the work of Pallotta et al., which is successful to determine the area of port and offshore platforms [30]. Even though both the works of Cerderio et al. and Pallotta et al. are able to identify stopping vessels within the port area or offshore platform with practical accuracy, they do not provide information on how long and where the vessels stop, whether they stop at the berthing location or within the anchorage area, and whether the anchorage area belongs to the POI.

A more recent study similar to the work of Cerderio et al. is published by Iphar et al., which proposes a method for the extraction, quantification, and qualification of port calls by using vessel position data of the AIS transmissions [31]. To extract port calls from the AIS tracks, they define a cut value for port proximity and a speed threshold that characterizes the beginning and end of a port call. The port proximity cut value is set to 925 m, and the speed threshold is 0.5 knots. While the speed threshold is a commonly accepted value regardless of location, the cut value for the port proximity could be region-dependent. For example, the port gate, where the port's pilot boards the entering ship or disembarks the departing ship, of Port Klang, Jakarta, and Surabaya are roughly 26 km, 7 km, and 41 km, respectively.

They also propose a technique for determining a vessel's berthing location by calculating the unweighted barycenter

of vessel locations during a port call for all points whose speed value is null. However, since the technique does not classify the identified port calls as berthing or anchoring ships, the computed barycenter could indicate the barycenter of a moored vessel or a ship at anchor. Thus, the identified berthing location could be either a berthing or an anchoring point.

Although their work is able to identify port calls with practical accuracy verified with the data declared by the UNCTAD and relevant port authorities, it does not explicitly provide the duration of the port calls to indicate the vessel turnaround time.

Based on the mentioned existing works, four tasks are required to be accomplished, and five features are necessary to transform raw AIS messages into meaningful indicators. The tasks are (1) the extraction of stopping tracks, (2) the estimation of port boundary, (3) the classification of the stopping tracks, and (4) the integration with other relevant sources of information. The features are (1) the number of port-call, (2) the container carrying capacity, (3) the number of services and companies, (4) the size of the largest ship, and (5) the number of other countries connected through direct liner shipping services. Then, there are three essential indicators for assessing port performance: (1) vessel turnaround time, (2) port waiting time, and (3) liner shipping connectivity. However, they missed the task of identifying the berthing location and anchorage area of the POI and how those places are connected. If the anchorage area is not connected to the POI, then the ships stopping at the area should not be included in the construction of the indicators.

In this paper, the tasks are performed using trajectory segmentation and stopping segment classification techniques, while the features will be generated by integrating the segmentation and classification results with an online vessel database consisting of more than 200,000 unique vessels [19]. Finally, the features and results of the tasks are employed to construct port performance indicators and compare them with widely accepted official statistics.

3 Proposed approach

This paper proposes a method to assess port performance by computing the port performance indicators based on the fusion of AIS-generated vessel trajectory segmentation and classification with countries' political administrative boundaries [20], vessels' attributes [19], and the point location of the ports of interest [21]. The approach incorporates five main steps: (1) trajectory segmentation, (2) stopping segment clustering and port boundary identification, (3) stopping segment classification, (4) port boundary verification, and (5) calculation of the port key performance indicators.

This five-step computation is preceded by an AIS data preprocessing to remove invalid records. Although rare, recorded AIS messages might have empty or invalid MMSI, latitude, longitude, heading, and COG values. Therefore, the preprocessing includes MMSI, latitude, longitude, heading, and COG filters. The complete filters are defined as follows.

- (1) $length(MMSI) = 9$, to select AIS messages with valid MMSI number. The reason is that the MMSI number should be unique and represent a unique ship. Non-valid MMSI may be used by more than one ship, or it could be testing equipment that does not represent a real ship [15].
- (2) $-90 \leq Latitude \leq 90$ and $-180 \leq Longitude \leq 180$, to sort out only the records with valid location coordinates.
- (3) $0 \leq heading \leq 360$ and $0 \leq COG \leq 360$, where COG is course over ground. It is to pick only the records with valid heading and COG values because both values are needed to calculate the discrepancy between COG and heading.
- (4) $70 \leq vesselTypeCode \leq 89$, to choose vessels of types cargo and tanker. Vessel of type cargo includes container, bulk carrier, general cargo, RO-RO, and special cargo ship. Vessel of type tanker consists of oil, gas, and chemical tanker. The ships of types cargo and tanker are selected because they are the merchant fleet that form the world's maritime transport [32, 33].
- (5) Time interval between two consecutive tracks $\delta t \leq 3600$ seconds or distance between two consecutive tracks $\delta d \leq 12$ nautical miles (nm). It is to avoid processing truncated trajectories. Any trajectories that do not meet the δt or δd threshold are ignored.

Ideally, δt should be the same as the AIS specifications, which have a 2 to 10-second transmission interval for ships moving faster than 3 knots. However, budget availability may limit the acquisition of full-resolution historical AIS datasets, particularly when they cover a wide area of multinational regions. This paper utilizes a historical AIS dataset with a 1-hour resolution purchased from Spire Global [34].

The rationale behind $\delta d \leq 12$ nm is that when $SOG = 12$ knots and the maximum $\delta t = 1$ hour, then the maximum $\delta d = 12$ nm. The 12 knots SOG is derived from the rules of the International Maritime Organization (IMO) [35] and the Maritime and Port Authority of Singapore [36] concerning the safe speed and the maximum allowed speed for vessels navigating the Straits of Malacca and Singapore. The work of Zhang et al. confirmed the implementation of the rules by revealing that ships mostly navigated the straits with $SOG \leq 12$ knots [37]. At the global level, Filipiak et al. divide the world in 33123 segments of $1^\circ \times 1^\circ$ and calculate the mean

SOG of tankers' traffic [38]. The mean is 11.63 knots, and the median is 12.53 knots.

3.1 Trajectory segmentation

The voyage trajectory of a vessel generated from AIS transmission data consists of stopping and underway tracks. Thus, the trajectory can be divided into stopping and underway segments.

Two thresholds are applied to separate the stopping segment from the underway one: (1) speed over ground (SOG) and (2) the discrepancy between course over ground (COG) and heading. Later, the discrepancy between COG and heading is denoted as $ch_discrepancy$.

The SOG threshold is 0.4 knots. It is determined based on seafarers' experience and the knowledge that SOG observed with GPS might not be reliable when it is less than 0.5 knots [39]. In other studies, Cerderio et al. applied speed below 0.5 knots threshold to construct moored and anchored vessel clusters [17], while Li et al. consider that at the hoteling state, ship speed is generally less than 0.5 knots [40]. In the state, a ship is usually fixed with cables to receive services such as loading and unloading, waiting at anchor, or other things.

The $ch_discrepancy$ feature is selected due to the distinctive nature of the underway and stopping vessels. Underway vessels move forward, so they typically do not experience a significant discrepancy between COG and heading. Meanwhile, stopping vessels, particularly vessels at anchor, often move sideways due to the force of wind and tidal stream, thus, they produce a considerably higher rate of $ch_discrepancy$.

Based on these two thresholds, the separation of stopping and underway segments of a voyage trajectory is carried out as follows.

- Step 1: Every trajectory T is defined as $T = \{p_0, p_1, p_2 \dots p_n\}$; p_k is a track position at timestamp t_k where $0 \leq k \leq n$. t_0 is the starting timestamp, and t_n is the end.
- Step 2: Traverse every trajectory T until the last track position p_n . If a track p_i with an $SOG \leq 0.4$ knots is found, then do: (1) Mark p_i as the start of a stopping segment, (2) mark p_0 as the start and p_{i-1} as the end of an underway segment, where $i > 0$, (3) continue traversing the T . If a track of $SOG > 0.4$ knots and $ch_discrepancy < 35$ is found, then do: (1) traverse backward to find the initial track p_j with $SOG > 0.4$ knots, (2) mark p_{j-1} as the end of the stopping segment.
- Step 3: Return to Step 2 and set p_j as the starting track position.

This procedure separates every stopping and underway segment of a vessel trajectory within the specified duration or length of the trajectory. The length of time from the start to the

Table 1 The six features thresholds to examine whether an at-anchor segment candidate is actually an at-anchor segment

Segment	<i>rothd_mean</i>	<i>dh24_sum</i>	<i>dh_std</i>	<i>d_seg</i> (yard)	<i>bb_area</i> (nmi ²)	<i>bb_lenwid_ratio</i>
At anchor	≥ 0.16	$\geq 600 \ \& \ < 3600$	$\geq 10 \ \& \ < 27$	$> 105 \ \& \ \leq 1200$	$> 0.0025 \ \& \ \leq 0.36$	$\geq 6/10 \ \& \ \leq 10/6$

end of every segment is defined as *segment_duration*. The *segment_duration* of a stopping segment represents how long a vessel stops, while *segment_duration* of an underway segment means how long a vessel navigates through the water. Since MMSI is the vessel's unique identity, each trajectory is identified with MMSI, and the separated stopping and underway segments are numbered sequentially. The sequence number combined with the MMSI defines the identity of the segments. So, each segment has a *segment_id*, formed by concatenating the MMSI and the sequence number.

3.2 Stopping trajectory classification to identify area and event within the POI

To calculate the performance indicators of the POI, first, the *de facto* area of the port should be identified. Then, the vessel activities within the area need to be recognized to explain the following: (1) where the vessels drop anchor, which is the anchorage area; (2) where they are moored, which is the berthing location, and (3) whether vessel movement exists between the anchorage and berthing locations. If vessel movement exists between the anchorage and berthing locations, these areas are related, as they belong to the same harbor.

The *de facto* area of the POI is estimated by spatially clustering the stopping segments. The logic is that if the stopping vessels' positions on a particular area of the earth's surface large enough to include multiple harbors are visualized on a flat map, it should be observable that the positions are spatially clustered or grouped by the area of the harbors where the vessels stopover. For this purpose, the DBSCAN clustering algorithm [29] is applied to the stopping segments. In this case, the Euclidean Distance is set to the *DistanceMetric* argument of the DBSCAN implementation.

The anchorage area and the berthing location of the POI are identified by classifying the stopping segments into at-anchor and moored segments. The water area filled with the at-anchor segments should be the anchorage area, while that occupied by the moored segments should be the berthing location. Two-step stopping segment classification is applied as follows.

First, the stopping segments are classified based on each segment's nearest distance to the port's land. If the distance is less than 210 m, it must be a moored segment. Otherwise, it is a candidate for an at-anchor segment. The Admiralty Manual of Navigation defines a *Safety Swinging Circle* (SSC) for

vessel at anchor [41]. The radius r of the SSC can be obtained by adding the length of ship (len_{ship}), the maximum amount of cable (max_{cable}), and a safety margin ($safemargin$) as in Eq. 1. Typically, ships have $max_{cable} = 275$ meters at the least. Suppose the $len_{ship} = 150$ meters and $safemargin = 275$ meters, then the SSC radius $r = 700$ meters. According to reference [41], it is usual to allow a least one cable (185 m) of safety margin. Therefore, the SSC radius r will always be greater than 210 m. In other words, there should not be any tanker or cargo ship anchoring within 210 m to the land.

$$r = len_{ship} + max_{cable} + safemargin \quad (1)$$

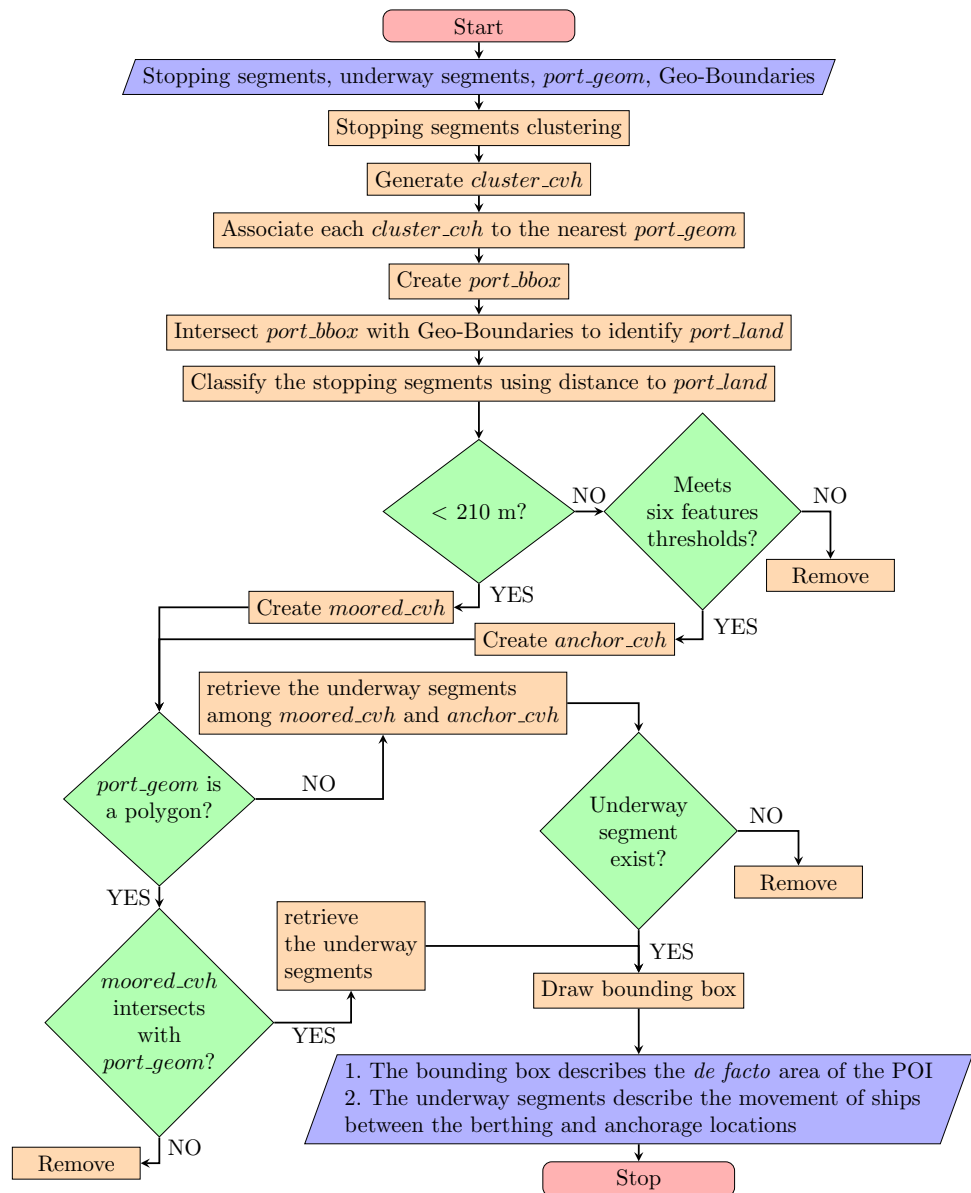
The 210 m threshold is determined by roughly considering AIS position accuracy [42], the length and width of tankers and cargo ships, the position of the antenna of the shipborne AIS, the accuracy of the map, and the influences of wind and tidal stream to the moored ship. The wind and tidal stream forces can move the ship tightly attached to or slightly away from the wharf.

Second, the candidate of the at-anchor segment is examined with the method proposed by Wijaya and Nakamura [43]. Based on the method, a smoothing procedure is applied to the candidate of the at-anchor segment, then the thresholds of six features are implemented to verify whether the candidate is an at-anchor segment. Those six features are: (1) *rothd_mean*, which is the average of the rate of heading change against the distance between two consecutive tracks, (2) *dh24_sum*: which is the sum of heading change per day, (3) *dh_std*: the standard deviation of heading change between two consecutive tracks, (4) *d_seg*: the segment diameter, (5) *bb_area*: the stopping segment's bounding box area, and (6) *bb_lenwid_ratio*: the bounding box's length and width ratio of the stopping segment. Table 1 presents the thresholds of the six features. If the candidate of the at-anchor segment meets the thresholds, then it is an at-anchor segment. Otherwise, it will be ignored as a noise.

These six features of the at-anchor segment are derived from the phenomenon that anchoring vessels experience a circular-like motion centered at the point where the anchor is dropped. This circular-like motion produces a significantly higher rate of heading change than the non-anchoring vessels and determines the bounding box dimension of an at-anchor segment.

Based on the heading change between two consecutive tracks, *rothd_mean*, *dh24_sum*, and *dh_std* are calculated. Meanwhile, *d_seg*, *bb_area*, and *bb_lenwid_ratio* are pro-

Fig. 1 The flowchart of the stopping trajectory classification to identify area and event within the POI



duced from the dimension of the at-anchor segment bounding box. The thresholds of these six features, presented in Table 1, are validated with AIS tracks of 1000 vessels recorded within one month.

Let $T = \{p_0, p_1, p_2 \dots p_n\}$; p_k is a track position at timestamp t_k where $0 \leq k \leq n$. ΔH_i and d_i are the heading change and distance between two consecutive tracks p_i and p_{i+1} , respectively. Then, $rothd_mean$ can be expressed as follows.

$$rothd_mean = \frac{1}{n-1} \sum_{i=0}^{n-1} \frac{\Delta H_i}{d_i} \quad (2)$$

After the berthing location and the anchorage area are identified, the underway segments acquired in the trajectory

segmentation are utilized to examine whether vessel movements exist between them. If at least one underway segment starts at the berthing location and ends at the anchorage area, or vice versa, then it implies that vessel movement exists between those two places.

The thorough procedure to classify the stopping segments and to identify areas and events within the POI is carried out in the following steps:

Step 1: Generate the convex hull of each cluster. Later, it is denoted as *cluster_cvh*.

Step 2: Associate each *cluster_cvh* to the nearest port. The port geographic location is expressed as a point or a polygon (when the official port boundary is publicly available) in longitude and latitude coor-

ordinates. It is denoted as *port_geom*. The distance between each *cluster_cvh* and each *port_geom* within the experiment geographic area is calculated. The *cluster_cvh* is associated to the nearest port when the distance between *cluster_cvh* and *port_geom* is less than 50 nautical miles. The *port_geom* data used in this experiment is based on the ports' point location of the World Port Index database [21] and the port boundary data made available by its respective authority. In this case, only the geometrical boundary of the Singapore port is publicly available [44]. If there is only one *cluster_cvh* associated with a *port_geom*, then the berthing location(s) and the anchorage area(s) of the POI should be located within the same *cluster_cvh*.

- Step 3: Create a bounding box *port_bbox* encompassing all of the *cluster_cvh* associated with the same *port_geom*.
- Step 4: Intersect the *port_bbox* with the Geo-Boundaries [20]. It selects the land area of the *port_bbox*. This should be the land area of the POI, denoted as *port_land*.
- Step 5: Classify the stopping segments, whether they are moored or at-anchor, based on the distance of each segment to the *port_land*. The distance is denoted as d_{land} . If $d_{land} < 210$ meters, then it is a moored segment, jump to Step 7. Else, it is an at-anchor segment candidate.
- Step 6: Examine the candidate of the at-anchor segment by applying the method described in the second step of the two-step stopping segment classification of this subsection. If the candidate meets the six feature thresholds, then it is an actual at-anchor segment. Else, ignore it as a noise.
- Step 7: For each *cluster_cvh*, that should contain moored and/or at-anchor segments, create the convex hull of these moored and/or at-anchor segments. The convex hulls of the moored and at-anchor segments are denoted *moored_cvh* and *anchor_cvh*, respectively. The *moored_cvh* estimates the berthing location, while *anchor_cvh* represents the estimated boundary of the anchorage area belonging to the POI.
- Step 8: If the *port_geom* is a polygon representing the geometrical boundary of the POI, evaluate whether *moored_cvh* intersects with the *port_geom*. Remove the one that does not intersect. Next, retrieve the underway segments that start from either *moored_cvh* or *anchor_cvh* to end at the *port_geom*, and vice versa with $segment_duration \leq 6$ hours. Jump to Step 10. Else, retrieve the underway segments that start from each *moored_cvh* or *anchor_cvh*

to end at the *moored_cvh*, and vice versa with $segment_duration \leq 6$ hours.

- Step 9: If a *moored_cvh* or *anchor_cvh* has at least one underway segment which start and/or end at it, then it belongs to the POI. Else, it belongs to another port.
- Step 10: Draw a bounding box encompassing all POI's *moored_cvh* and *anchor_cvh*. This bounding box estimates the *de facto* rectangular boundary of the POI. Meanwhile, the retrieved underway segments imply the vessel's movement between the berthing location and the anchorage area. Later, these underway segments and the movement are denoted as maneuvering segments and maneuvering, respectively.

The final results of this procedure are: (1) the estimated *de facto* area of the POI, (2) the berthing location and anchorage area that belong to the POI, and (3) the berthing, anchoring, and maneuvering events within the POI's boundary. Figure 1 summarizes the whole process of achieving the results.

3.3 Generating port performance indicators

The key indicators to measure port performance defined by the UNCTAD have been introduced in Sect. 2, of which vessel turnaround time or vessel time in port is regarded as the most essential. This study calculates the vessel turnaround time based on berthing duration, retrieved from the duration of the moored segment within the berthing location, and maneuvering duration, which is the duration of the maneuvering segment. Equation 3 defines the calculation explicitly, where *turnaround* is the turnaround time, *berth* is for the berthing duration, and *maneuver* is the maneuvering duration.

$$turnaround = berth + maneuver \quad (3)$$

In addition to the vessel turnaround time, this study also produces other indicators that could be necessary for port users and administrators, such as (1) the time spent at anchor, (2) the size range of vessels in gross tonnage and dimension, (3) the deadweight of container ships, and (4) the connectivity of the POI with other ports and with the international shipping companies.

First, the time spent at the anchorage area can be used to measure the ship waiting duration outside the port. This indicator is necessary for port administrator to optimize the port organization and management plan to reduce congestion, and it is crucial for ships to plan their voyage and reduce costs [45].

Second, the range of vessel size, the deadweight, and the connectivity of the POI with other ports and shipping companies can be useful in interpreting the Liner Shipping

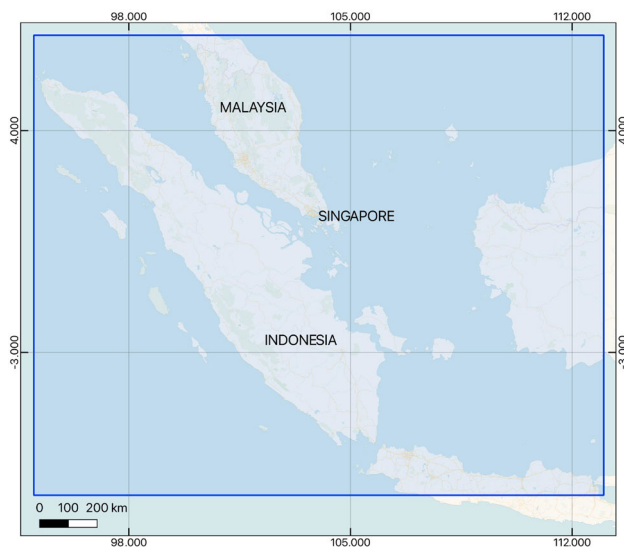


Fig. 2 The area within the blue rectangle is the experiment's area of interest. Basemap © CARTO © OpenMapTiles © OpenStreetMap contributors

Connectivity, which represents the port connectivity with the global shipping network [26]. These indicators are produced by integrating the stopping segment classification and the area identification results in Sect. 3.2 with the vessel database available at BalticShipping.com [19]. The database contains detailed vessel attributes, such as vessel type, gross tonnage, deadweight, dimension, home port, and owner (company or individual). It can be queried with the International Maritime Organization (IMO) number included in the AIS messages.

UNCTAD provides the Liner Shipping Connectivity Index (LSCI), which indicates a country's position within global liner shipping networks. It is calculated based on the following variables: (1) the number of ship calls, (2) their container carrying capacity, (3) the number of services and companies, (4) the size of the largest ship, and (5) the number of other countries connected through direct liner shipping services [27]. The proposed method in this paper facilitates the fulfillment of those variables. The number of ship calls can be derived from the number of stopping segments, while their container carrying capacity is represented by their deadweight. The size range of vessels in gross tonnage provides the size of the largest ship of the POI. Finally, the number of other countries connected through direct liner shipping can be inferred from the POI's connectivity with other ports and international shipping companies.

4 Experiment and discussion

This section describes the implementation and evaluation of the proposed method based on real-world historical AIS data of vessels navigating through the southern part of the South

China Sea, western Indonesia, Singapore, and Malaysian waters. Figure 2 depicts the experiment's area of interest, roughly 3,230,663.98 km².

The historical AIS data were purchased from Spire Global [34] through a local representative. They were recorded between 1st July 2021 and 28th February 2022 within the specified experiment's area of interest. The AIS dataset is enriched with the vessel database at Balticshipping.com [19]. The database is available online and can be queried with vessel's name, IMO number, and ship type to get detailed information about the ship of interest.

The experiment is conducted by following the five main steps described in Sect. 3.

4.1 AIS data preprocessing

The one-month historical AIS data of 1st–31st July 2021 are filtered to retrieve all MMSIs of valid AIS messages of cargo ships (container, bulk carrier, general cargo, RO–RO, special cargo) and tankers (oil, gas, other). The following filters are applied: $length(MMSI) = 9$, $-90 \leq Latitude \leq 90$, $-180 \leq Longitude \leq 180$, and $70 \leq vesselTypeCode \leq 89$.

The filtering results in 6950 unique MMSIs, of which 4182 MMSIs are used as the keys to fetch one-month historical AIS messages of 1st–31st July 2021. This consists of 3,514,126 AIS messages. The remaining 2768 MMSIs are the keys to retrieve eight-month historical AIS messages between 1st July 2021 and 28th February 2022 that contain 15,955,795 AIS records. Thus, this experiment processes 19,469,921 AIS messages in total.

4.2 Trajectory segmentation

The stopping and underway segments separation algorithm is implemented on the one-month and eight-month AIS messages. The segmentation is conducted per MMSI. The AIS messages belonging to an MMSI are sorted by timestamp in ascending mode. This is to construct the vessel trajectory within the specified duration (one month and eight months). The implementation of the trajectory segmentation algorithm divides the trajectory into stopping and underway segments. Each segment is given a *segment_id* and *segment_duration* attributes as described in Sect. 3.1. The *segment_duration* is the duration between the segment's starting and ending timestamp. For each segment, the time interval δt and the distance δd between two consecutive tracks are calculated. Any segment that has $\delta t > 3600$ seconds or $\delta d > 12$ nm is removed as defined in the AIS data preprocessing filters in Sect. 3. Finally, for each valid segment, a geometry of type LineString is constructed with the segment's longitude and latitude coordinates. The LineString

geometry of each segment is used to evaluate the experiment result and visualize the segment on a relevant map.

The evaluation of the experiment results uses a considerably smaller geographic area within the experiment's area of interest. It is to enable a human-readable visualization so that the experiment results can also be visually evaluated. For this purpose, the AIS data within the boundary of the Singapore port is selected for two reasons: (1) Singapore port's official boundary is publicly available, and (2) the data occupies roughly 86% of the whole experiment AIS data. Additionally, Singapore port is considered the port with the second highest connectivity index in the world [46], and it is the second-busiest port in the world in terms of total shipping tonnage [37]. It can be considered the most significant harbor within the experiment's area of interest. Figure 3 illustrates the geographic area for evaluating the trajectory segmentation results.

The segmentation result is evaluated based on the following common knowledge:

1. Stopping trajectories may exist within either the port limit or at the OPL (Outside Port Limit). A ship is moored only at a designated berthing facility within the port. However, she may drop anchor at the anchorage area within or outside the port limit.
2. Stopping trajectory should not exist within the fairways as any normally operating ship shall not stop within the fairways.
3. Stopping trajectory should not exist within the TSS (Traffic Separation Scheme). The 1972 Convention on the International Regulations for Preventing Collisions at Sea (COLREGS 1972) regulates that a vessel using a TSS shall proceed in the appropriate traffic lane in the general direction of traffic flow for that lane and, so far practicable, avoid anchoring in the TSS or in areas near its terminations [47].
4. Underway trajectory should intersect with either the TSS, the fairways, or both. Vessels navigating to or out of the Singapore port or transiting the Singapore Strait will surely pass through either the fairways, the TSS, or both.
5. Underway trajectory should cross the port limit as they enter or exit the port. The Singapore port area is delineated with a polygon. Thus, any ship entering or departing the port will cross the imaginary line of the polygon.

To ensure the quality of the experiment datasets, both the one-month and the eight-month historical AIS messages are filtered to meet the following requirements: (1) the length of each stopping and each underway segment must not be less than 6 h, and (2) each underway segment consists of at least 12 points, and the distance between two adjacent points is not more than 3000 yards (2778 m).

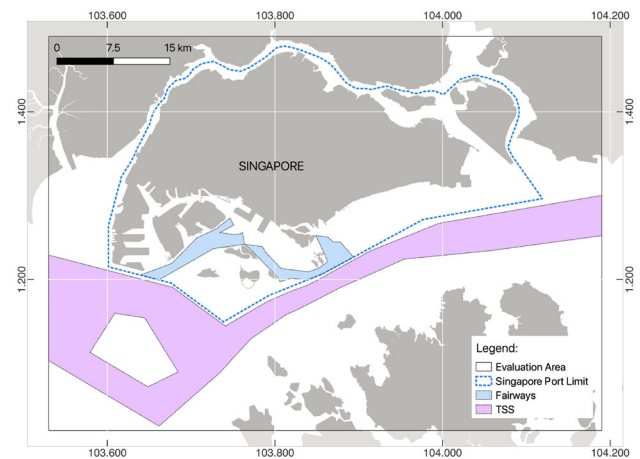


Fig. 3 The area within the black rectangle is the geographical scope for the evaluation of trajectory segmentation results. Basemap adopts geoBoundaries [20]

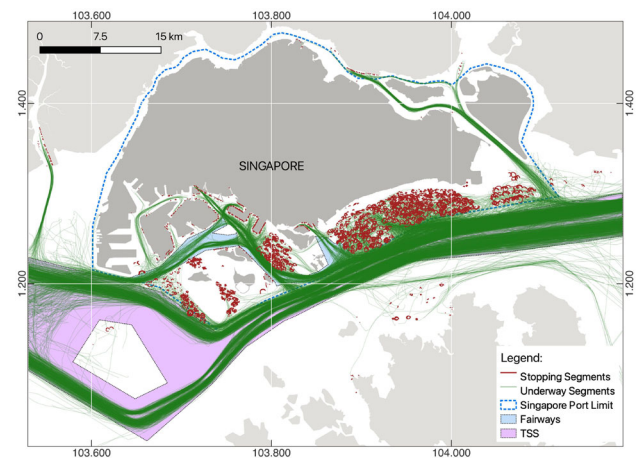


Fig. 4 The Trajectory segmentation results of the one-month AIS dataset. The AIS generated trajectories are divided into stopping and underway segments, visualized with red and green lines respectively. Basemap adopts geoBoundaries [20]

The trajectory segmentation experiment on the one-month AIS dataset generates 3509 stopping and 3488 underway segments. Figure 4 shows the segmentation results. The evaluation verifies 0.98 accuracy, with 4 and 105 segments falsely classified as stopping and underway segments, respectively. These falsely classified segments are illustrated in Figs. 5 and 6. The evaluation ruled that the four stopping segments were false because, according to COLREGS 1972, vessels shall not stop in the TSS or in areas near its terminations. However, when the four stopping segments are visually evaluated as in Fig. 5, they most likely depict vessels at anchor instead of underway. So, the four stopping segments should be evaluated as true. Likewise, the evaluation assessed the 105 underway segments as false due to the assumption that underway trajectories should intersect with either the TSS or fairways or both and should cross the port limit when the

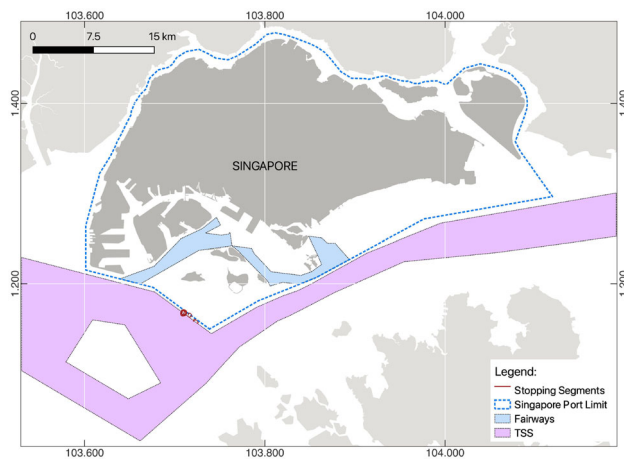


Fig. 5 The stopping segments are visualized in red near the coordinates (103.700, 1.150). The evaluation determined the segmentation results as false because, according to COLREGS 1972, vessels shall not stop in the TSS or areas near its terminations. Basemap adopts geoBoundaries [20]

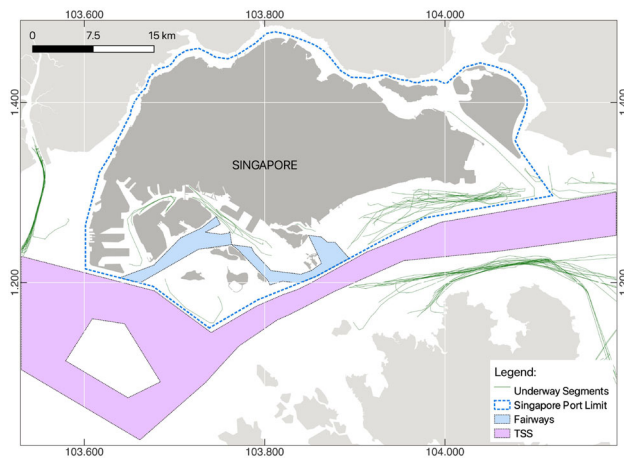


Fig. 6 The underway segments are visualized in green. The evaluation weighed up the segmentation results as false due to the assumption that underway trajectories should intersect with either the TSS or fairways or both and should cross the port limit when the associated vessel enter or exit the port. Basemap adopts geoBoundaries [20]

associated vessel enters or exits the port. However, based on the visual observation as in Fig. 6, the trajectories are most likely to represent underway vessels rather than stopping. Thus, these 105 underway segments should also be evaluated as true. It means that this experiment's results reached nearly 100% accuracy. Table 2 shows the complete evaluation metrics of the experiment results.

Likewise, in line with the experiment on the one-month AIS dataset, the trajectory segmentation of the eight-month AIS dataset also demonstrated desirable results with 0.99 accuracy. It yielded 19250 stopping and 4632 underway segments, as illustrated in Fig. 7. The evaluation revealed that 37 stopping and 124 underway segments are falsely classified.

Table 2 The evaluation metrics of the trajectory segmentation experiment on the one month AIS dataset

Trajectory segmentation	Precision	Recall	F1-score
Stopping segments ¹	0.99	0.97	0.98
Underway segments ²	0.97	0.99	0.98

¹Four false stopping segments out of 3509

²105 false underway segments out of 3488

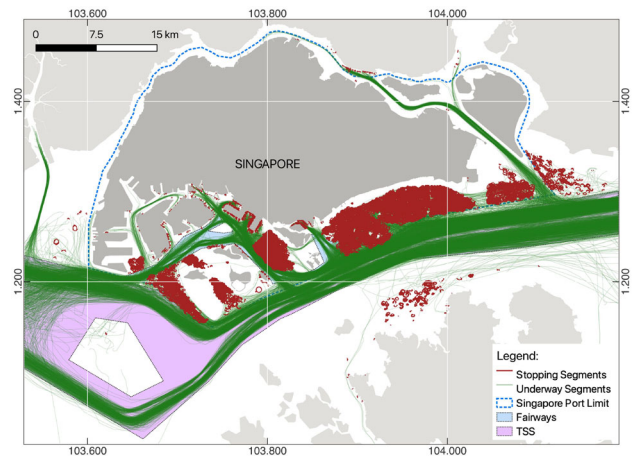


Fig. 7 The Trajectory segmentation results of the eight months AIS dataset. The AIS generated trajectories are divided into stopping and underway segments, visualized with red and green lines respectively. Basemap adopts geoBoundaries [20]

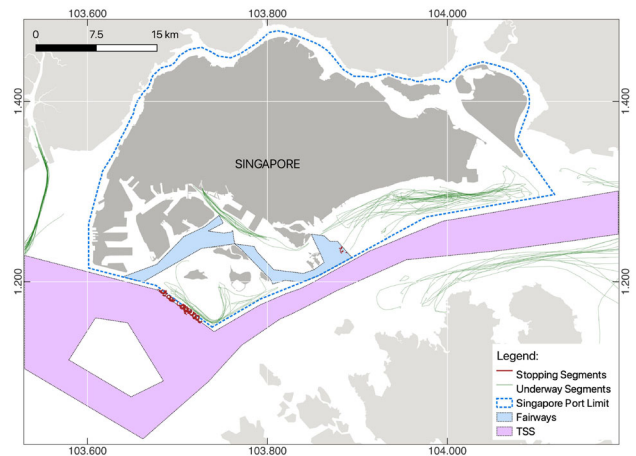


Fig. 8 The stopping segments are visualized in red, while the underways are in green. The evaluation weighed up these segmentation results as false. Basemap adopts geoBoundaries [20]

Some of the 37 stopping segments intersect with the TSS, while the others are within the fairways as depicted in Fig. 8. Therefore, they are evaluated as false-stopping segments. However, the visualization indicates that the segments are not likely to be the trajectories of underway vessels. In the same way, the 124 underway segments intersect with neither the TSS nor the fairways, nor do they cross the port limit

Table 3 The evaluation metrics of the trajectory segmentation experiment on the eight months AIS dataset

Trajectory segmentation	Precision	Recall	F1-score
Stopping segments ¹	0.99	0.99	0.99
Underway segments ²	0.97	0.99	0.98

¹ 37 false stopping segments out of 19250

² 124 false underway segments out of 4632

as shown in Fig. 8. Consequently, they are evaluated as false underway segments, although their visualization implies that they should be the trajectories of underway vessels. These findings suggest that the accuracy of this experiment results is roughly 100%. Table 3 shows the complete evaluation metrics of the experiment results.

The experiments' results on the one-month and eight-month AIS datasets verify the efficacy of the proposed vessel trajectory segmentation method. It demonstrates consistent results in relatively short (one month) and long (eight months) historical AIS data. The long historical AIS data are recorded over several seasons, which means that the vessels transmitting the AIS messages experienced significant changes in weather and other environmental conditions.

In addition, both results remarkably outperformed the existing method proposed by Yan et al., described in Sect. 2. They yielded 0.94 accuracies according to the experiment on a dataset of 359,322 AIS records reported by 70 vessels.

4.3 Stopping trajectory classification to identify area and event within the POI

This experiment processes all of the stopping and underway segments yielded from the trajectory segmentation experiments on both the one-month and the eight-month AIS datasets.

It selects three significant ports from three different countries as the ports of interest: the port of Singapore, Jakarta (Indonesia), and Port Klang (Malaysia). Each of these ports has different features. Singapore Port is located between the territorial waters of Indonesia and Malaysia and is surrounded by narrow straits. The Jakarta port is located in Indonesia's territorial waters and is directly connected to the open sea. Port Klang is located in Malaysia's territorial waters and is connected to the open sea by two narrow straits.

The clustering of the stopping segments, which are 29712 segments, results in 77 clusters. There are 62 segments that do not belong to any clusters categorized as noises. The DBSCAN's parameters are set as follows: $\epsilon = 0.015$, $MinPts = 2$, and $DistanceMetric = EuclideanDistance$. Since the distance unit of the Longitude and Latitude coordinates is in degree, the ϵ parameter value is equal to 0.9 nautical miles or 1,800 yards. A con-

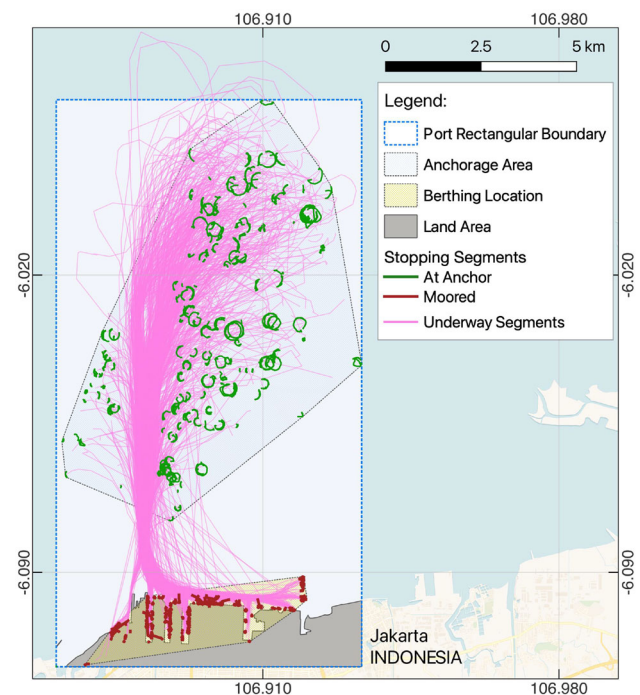


Fig. 9 The classification of areas and events within the port of Jakarta. It has one anchorage area and one berthing location. The visualization of the underway segments shows the vessels' movement between the anchorage area and the berthing location. Basemap © CARTO © OpenMapTiles © OpenStreetMap contributors

vex hull *cluster_cvh* is created for each stopping segment cluster, resulting in 77 *cluster_cvh*. Every *cluster_cvh* is associated with the nearest *port_geom* of the ports within the experiment's area of interest. The *port_geom* of the ports are obtained from the World Port Index database [21].

The port of Jakarta has four associated *cluster_cvhs*. However, two *cluster_cvhs* are removed upon conducting the stopping segment classification procedure, described in Sect. 3.2, illustrated in Fig. 1. The remaining two *cluster_cvhs* are a *moored_cvh* and an *anchor_cvh*. The bounding box encompassing the two *cluster_cvhs* estimates the *de facto* boundary of the port of Jakarta. The *moored_cvh* and *anchor_cvh* represent the berthing location and anchorage area of the port, respectively. The connection between those places is proven by underway segments that start and end at both *moored_cvh* and *anchor_cvh*, illustrated in Fig. 9.

In Step 6 of the procedure, each candidate of the at-anchor segment is examined to determine whether it has the characteristics of an at-anchor trajectory. Of the 170 candidates, 161, or 95%, passed the thresholds.

There are seven *cluster_cvhs* associated to the *port_geom* of the Port Klang. They consist of three *moored_cvhs* and four *anchor_cvhs*. None of them is removed during the execution of the stopping segment classification procedure.

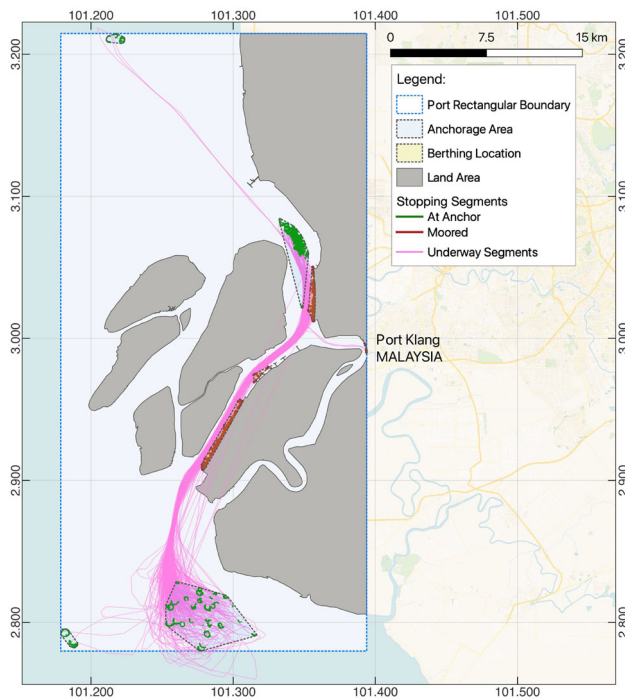


Fig. 10 The classification of areas and events within the Port Klang, Malaysia. It has four anchorage areas and three berthing locations. The visualization of the underway segments demonstrates that vessels moved among those anchorage areas and berthing locations. Basemap © CARTO © OpenMapTiles © OpenStreetMap contributors

Their connections are indicated by the underway segments that start and end amongst them. Thus, the *de facto* area of the Port Klang is estimated with four anchorage areas and three berthing locations. It corresponds to the fact that Port Klang has three berthing locations: the North, the South, and the West port. Figure 10 depicts the experiment results. At Step 6 of the procedure, 97% of the candidates, 116 segments of 119, matched the characteristics of the at-anchor trajectory.

The existing method of stopping segment classification, proposed in the work of Yan et al., demonstrated an accuracy of 93% on the experiment of 202 stopping segments [16]. Here, the proposed approach in this paper is proven to make a significant improvement.

The experiment on the port of Singapore is slightly different from those of Port Klang and Jakarta since the official boundary of the Singapore port is publicly available [44]. In this case, the official port boundary is used as the *port_geom* instead of using the data from the World Port Index database [21]. It has 22 associated *cluster_cvhs*. Four of them are removed during the execution of the stopping segment classification procedure. Thus, the *de facto* boundary of the Singapore port is estimated with the bounding box encompassing 18 *cluster_cvhs*. It is much greater than its official boundary.

Normally, a port has anchorage areas within the same territorial waters as the port. In these areas, vessels may drop anchor before entering and/or after departing from the port. However, the Singapore port is quite unique in this case. It is not only connected to the anchorage areas within the Singapore territory but also to those within the territorial waters of other countries [48–50]. The experiment results on the port of Singapore reveal this situation as depicted in Fig. 11.

To sum up, the experiments produced three expected results: (1) reveal the *de facto* boundaries of the three ports in the experiments, (2) identify the berthing locations and the anchorage areas associated with the ports, and (3) visually describe each port's berthing, anchoring, and maneuvering events.

While the existing study identifies stopping segments and determines the location point of the POI [16], this study makes a further improved approach by identifying the *de facto* boundary of the port and the zoning of the area: berthing location and anchorage area.

4.4 Port performance indicators and statistics

The following experiments involve the construction of three port performance indicators, which have already been recognized by existing publications.

4.4.1 Vessel time in port

This experiment calculated the median of moored segment duration and maneuvering duration that belong to container ships. The median moored segment durations in Jakarta, Port Klang, and Singapore ports are 1.04, 0.95, and 0.92 days, respectively, while the median maneuvering durations are 1.19, 1.58, and 0.92 h. By applying the Eq. 3, the median turnaround time or time in port of the three ports is 1.09, 1.02, and 0.96 days.

Here, the median time in port of the container vessels in Indonesia (Jakarta), Malaysia (Port Klang), and Singapore closely match the data of the UNCTAD [32, 51]. Table 4 compares the experiment results with the statistics of the UNCTAD. If the UNCTAD's statistics are considered to be the ground truth figure, then the rate of error of the experiment results can be calculated. For this purpose, the harmonic mean formula is adopted. The harmonic mean H of n numbers $x_1, x_2, x_3, \dots, x_n$ is given by Eq. 4. By applying the formula, the harmonic mean of the error rate is 2.9%. The existing study, which also calculates port performance indicators, does not provide an evaluation result for the median time in port [5]. Instead, it validates the number of ship arrivals in the POI with 3–4% of absolute percentage errors.

$$H = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_n}} \quad (4)$$

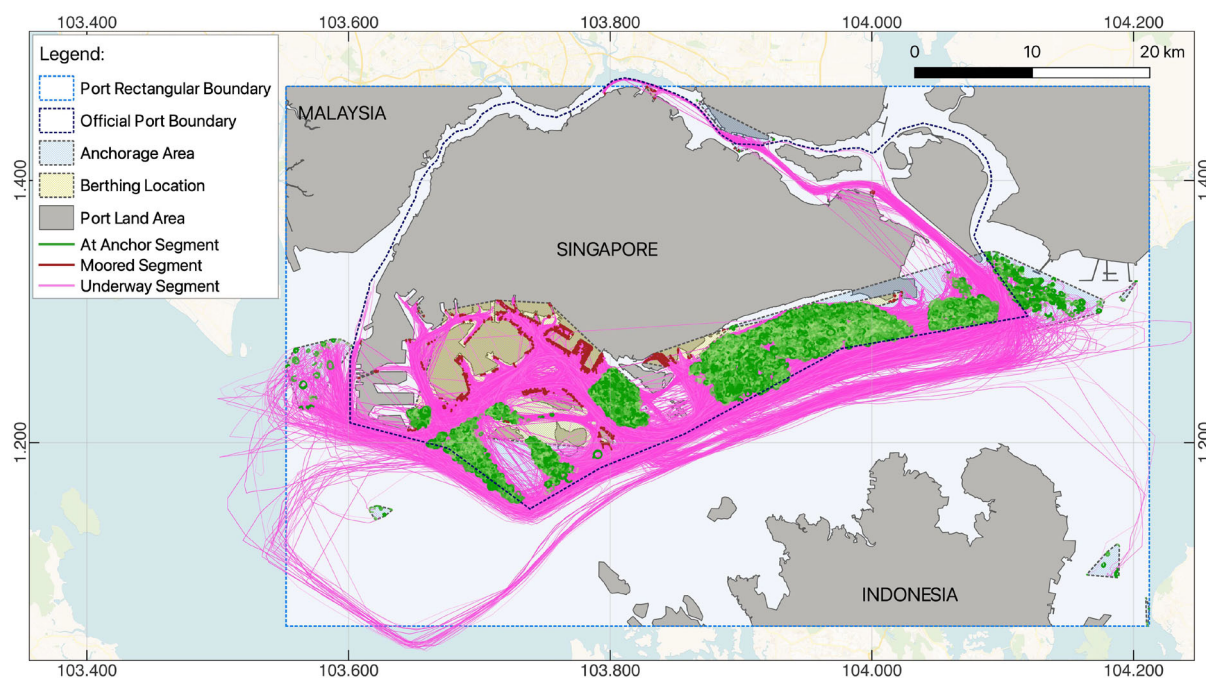


Fig. 11 Identifying areas and events within the port of Singapore. It has a publicly available official port limit, delineated by the dark blue dashed line. The visualization of the underway segments shows that vessels are moving among the berthing locations and anchorage areas, which are also located outside of the port limit. Some of these anchorage areas are even within the territorial waters of the neighboring countries.

This means that the geographic area occupied by the vessels' activities of the Singapore port is much greater than its official boundary. The port rectangular boundary outlined with the dodger blue dashed line depicts the area. Basemap © CARTO © OpenMapTiles © OpenStreetMap contributors

Table 4 Time spent in port (days) of container ships in 2021

Country	UNCTAD (median)	This study (median)	Error Rate ¹ (%)
Indonesia	1.06	1.09	2.8
Malaysia	1.00	1.02	2.0
Singapore	1.03	0.96	6.8

¹The harmonic mean of the error rate = 2.9%

UNCTAD produced the value of the vessel time in port indicator based on the data accumulated from various sources: satellite tracking of vessels, shipping schedules, and port information platforms. Meanwhile, this study yielded an almost identical value by merely utilizing AIS data of relevant space and time.

4.4.2 Waiting duration at the anchorage area

In addition to the port turnaround time, this study also produces other indicators that could be necessary for port users and administrators, such as (1) the time spent at anchor, (2) the size range of vessels in gross tonnage and dimension, and (3) the connectivity of the POI with other ports and with the international shipping companies.

The time spent at the anchorage area can be used to measure the ship waiting duration outside the port. This indicator

is necessary for port administrators to optimize the port organization and management plan to reduce congestion, and it is crucial for ships to plan their voyage and reduce costs [45]. Table 5 compares the vessel at anchor duration for all ship types at the port of Jakarta (Indonesia), Port Klang (Malaysia), and the port of Singapore. Although Singapore port served outstandingly more ships than both Port Klang and the port of Jakarta, it is most likely less congested indicated by the shorter anchoring duration.

In Singapore, the anchoring durations of vessels of types tanker and cargo are mostly within 24 h. The Oil tanker has the highest number, and several of them have the most extended anchoring duration, which is between 120 and 144 h (Fig. 12). Port Klang has similar characteristics to Singapore, illustrated in Fig. 13. However, the port of Jakarta demonstrated different features, depicted in Fig. 14. All vessels of subtype RO-RO of type cargo have an anchoring duration

Table 5 Anchoring duration (days) of tanker and cargo ships in 2021

Country	Median	Mean	Number of ships
Indonesia	1.01	1.15	613
Malaysia	0.68	0.84	760
Singapore	0.51	0.84	25684

of 24–48 h. The anchorage area is mainly populated by container ships, while bulk carriers have the longest anchoring duration of 120 to 144 h.

Figures 12, 13, and 14 present detailed subtypes of vessels that are not available in the AIS datasets. It is achievable by enriching the AIS datasets with the vessel database at BalticShipping.com [19].

4.4.3 Port connectivity

Line shipping connectivity represents the port connectivity with the global shipping network [26]. UNCTAD provides the liner shipping connectivity index (LSCI), which indicates a country's position within global liner shipping networks. It is calculated based on the following variables: (1) the number of ship calls, (2) their container carrying capacity, (3) the number of services and companies, (4) the size of the largest ship, and (5) the number of other countries connected through direct liner shipping services [27].

The values of those variables can be derived from the experiment results combined with the vessel database at BalticShipping.com [19]. In addition to the vessel static information contained within the experiment AIS data, the vessel database provides detailed information such as the vessel's gross tonnage, deadweight, ship dimensions (length and width), home port, owner, vessel type, and former names. The container carrying capacity can be obtained from the deadweight of container ships, and the number of ship calls of this type can be inferred from the number of stopping segments (moored and at-anchor) of container ships within the POI. The maximum gross tonnage indicates the size of the largest ship. The information about ship owner, which refers to the company owning the ship, and ship home port (overseas) imply the shipping connection between the POI with other countries.

Table 6 compares the connectivity of Singapore port, Port Klang (Malaysia), and the port of Jakarta (Indonesia). Based on the figures demonstrated by the three ports, Singapore is best connected to the global liner shipping network, followed by Malaysia and Indonesia. It agrees with the data presented in reference [27].

5 Conclusion

A consistent and reliable port performance is required to ensure efficiency in maritime transport, which establishes the foundation of the international trade and manufacturing supply chain. The Automatic Identification System (AIS), with its global coverage and straightforward accessibility features, has the capacity to be the primary source of maritime data. However, transforming the raw AIS messages into relevant metrics for constructing port performance indicators is undoubtedly challenging work. Even though AIS messages comprise the transmitting vessels' static, dynamic, and voyage-related information, they still need to be enriched with relevant information, such as a detailed vessel registry database and geographic information. Based on the existing literature, the tasks in exploiting AIS data to yield relevant outcomes include extracting and classifying stopping tracks, estimating port boundaries, and integrating the results with other relevant sources of information.

This study proposes a comprehensive technique to estimate port boundaries, identify the berthing location and anchorage area associated with the port, and construct port performance indicators based on AIS-generated vessel trajectories enriched with geographic information and vessel attributes.

The approach involves segmenting vessel trajectory into stopping and underway segments and classifying the stopping segments into moored and at-anchor segments. By integrating the trajectory segmentation and classification results with geographic information, the method estimates port boundary and identifies the port's berthing location and anchorage area. It also explains the connection between those places by examining whether vessel maneuvering exists between them. Finally, the method constructs port performance indicators by measuring the berthing, anchoring, and maneuvering duration and associating vessel identity with a vessel database. Vessel turnaround time, known as one of the most important indicators of port performance, is yielded from the sum of berthing and maneuvering duration. The entire process is a step-by-step, thorough procedure coupled with evaluations to transform raw AIS data into meaningful values.

To verify the practicality of the proposed approach, experiments are carried out on more than 19 million AIS messages recorded within eight months over a region of more than 3 million km² that includes one of the busiest shipping routes on earth. The trajectory segmentation demonstrated 98–99% accuracy, while the classification achieved 95–97%. The estimation of vessel turnaround time closely matched the figure provided by the UNCTAD, with an error rate of 2.9%. These experiment results imply the practicality of the proposed approach in contributing to the real-world scenario.

Fig. 12 The anchoring duration (hours) of types Cargo and Tanker in Singapore port

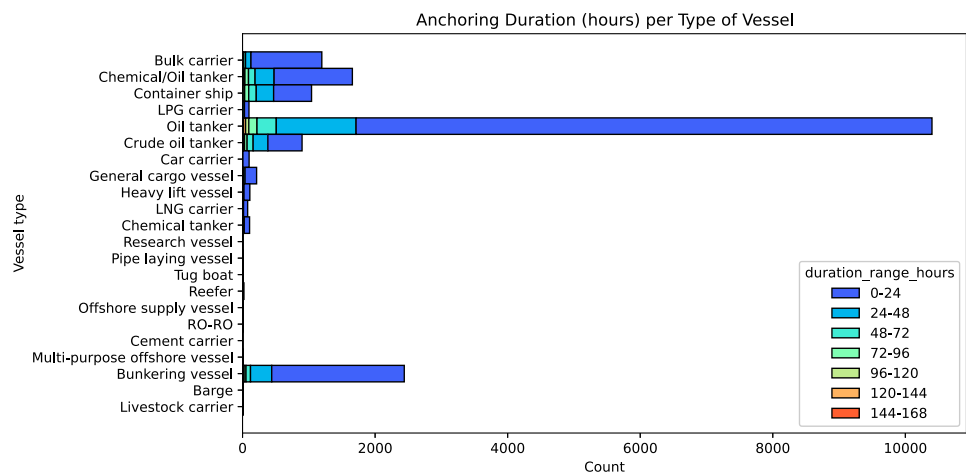


Fig. 13 The anchoring duration (hours) of types Cargo and Tanker in port Klang

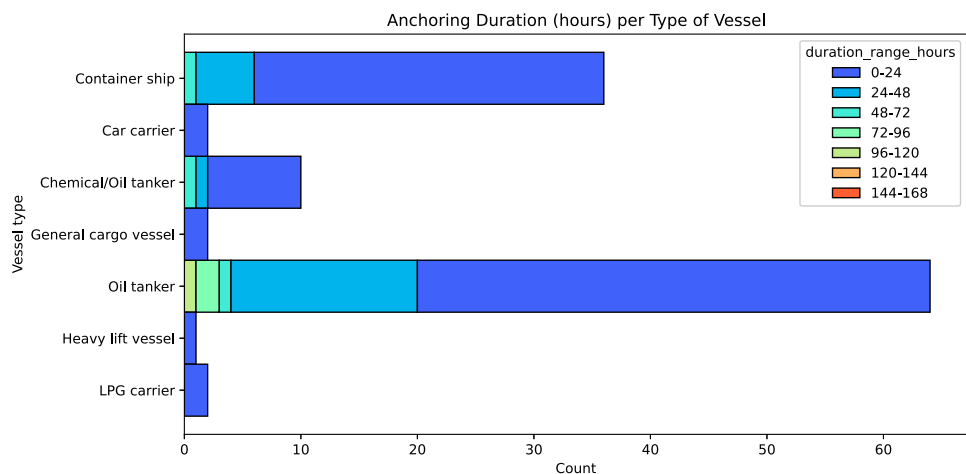


Fig. 14 The anchoring duration (hours) of types Cargo and Tanker in the port of Jakarta

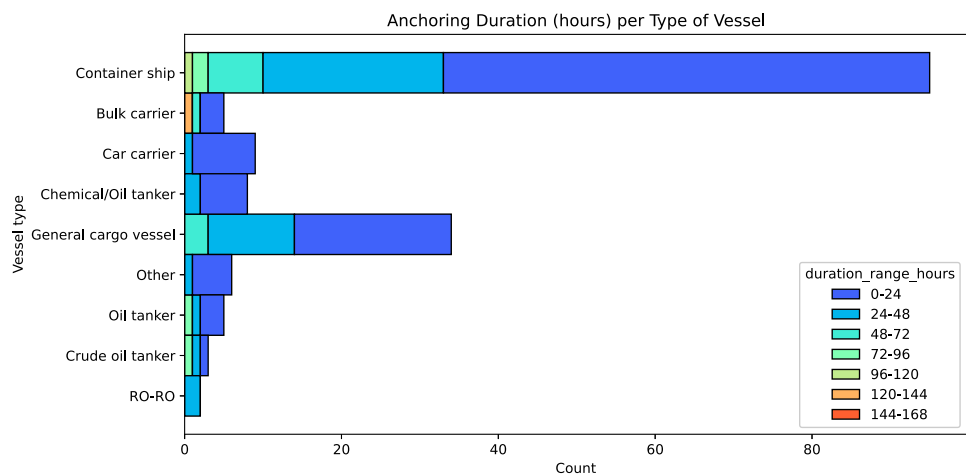


Table 6 Port connectivity comparison

Country	Ship call (container ship)	Largest ship gross tonnage (ton)	Deadweight mean (ton)	Connected overseas port	Connected ship-ping company
Indonesia	314	66332.0	25776.23	33	142
Malaysia	503	219277.0	45365.49	37	165
Singapore	3299	232618.0	52999.94	117	1175

The result is significant because constructing indicators to assess port performance has proven difficult due to issues with the quality, consistency, availability, and accessibility of the official data. The process is often time-consuming, limiting the opportunity for stakeholders to respond in a timely manner to emerging trends. With the AIS data's real-time nature, global coverage, and straightforward accessibility, the proposed method in this paper is potent for delivering the indicators ahead of the official statistic dissemination.

This paper contributes in the following ways. First, provides an accurate technique to separate stopping and underway segments from vessel trajectories. This facilitates the analysis of vessel activities in port areas and the identification of unusual behavior in vessel movement patterns at sea. Second, estimates the *de facto* boundary and zoning of the POI based on the observed vessel activities in the area. Maritime space management can employ the estimation to assess how well activities observed align with existing zoning regulations. Third, proposes a transparent and timely approach to constructing port performance indicators, thereby benefiting the global maritime industries.

Despite the achievements, this study identifies limitations and future works for further development. The vessel turnaround time is calculated based on the duration of berthing and maneuvering between the berthing and anchoring locations. The distance between the berthing and anchoring locations is assumed to represent the distance from the port entrance (or pilot boarding area) to the berthing location. This assumption is used because the actual location of the port's entrance is not known. In addition, the detailed information concerning the ship's identity and specifications is retrieved from an online database available at BalticShipping.com, but it does not contain all vessels in the experiment datasets. Thus, to achieve a more realistic representation of the real-world scenario, future works should include a method to identify the location of the port's entrance and utilize databases that comprise all vessels within the area of interest. Another limitation is that this study assumes all berthing events are related to ship loading/offloading and other activities concomitant to the process. However, a ship moored at a shipyard is not likely to engage in cargo loading/offloading activities. Therefore, to improve the practicality of the proposed method, a classification technique to identify the type of berthing events needs to be formulated. Finally, a more advanced approach should be employed to define the more natural *de facto* boundary of the POI instead of simply drawing a bounding box.

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